

The Semantic Arms Race: From Stylometric Fragility to Agentic Defense in AI-Generated Spam

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Abstract

The democratization of Large Language Models (LLMs) has fundamentally altered the economics and mechanics of phishing, transitioning the threat landscape from syntax-based spam to semantic-based "industrialized mimicry." Recent empirical measurements indicate that over 51% of spam is now AI-generated, utilizing adversarial paraphrasing to evade traditional signature-based filters. This essay critically analyzes the current "arms race" between generative evasion and stylometric detection. We evaluate the efficacy of stylometry (Opara et al.) as a defensive mechanism and contrast it with the fragility of human-in-the-loop systems, which are prone to automation bias (Flores et al.). We conclude that while stylometry provides a temporary defensive advantage, robust security requires a paradigm shift toward "Agentic Defense", systems that verify intent through external context rather than internal syntax.

1 Introduction

Historically, the defense against phishing and spam relied on the "cost of quality." For an attacker, crafting a persuasive, context-aware, and grammatically perfect email required significant human effort. Consequently, spam was typically high-volume but low-quality, allowing defenders to rely on simple heuristics: typos, broken grammar, and known malicious signatures were high-fidelity signals of fraud.

The advent of Large Language Models (LLMs) has inverted this dynamic. Tools like GPT-4 allow adversaries to generate high-quality, polymorphic spear-phishing campaigns at negligible cost. Heiding et al. [3] demonstrated that LLMs, when combined with psychological principles like the "V-Triad" (Visceral, Verification, Vulnerability), can achieve

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click-through rates as high as 81%, compared to roughly 20% for traditional phishing baselines. This capability allows low-resource adversaries to operate with the sophistication previously reserved for state-sponsored actors.

This essay defines the current "arms race" between evasion and detection. First, we analyze the *Offensive Paradigm* (Section 2), specifically how attackers use "Adversarial Paraphrasing" to scrub AI artifacts from text. Second, we examine the *Defensive Paradigm* (Section 3), focusing on Opara et al.'s work on stylometric detection and its limitations. Third, we critique the failure of Human-AI collaboration (Section 4) due to cognitive offloading. Finally, we argue that the future of defense lies not in analyzing text, but in "Explainable Agentic AI" that verifies external truth (Section 5).

2 The Offensive Paradigm: Industrialized Mimicry

The core threat of LLM-based spam lies in the decoupling of content from form. Traditional spam filters rely heavily on static signatures, specific URL structures, hash values, or repetitive grammatical errors, to identify malicious campaigns. LLMs dismantle this paradigm by enabling *Adversarial Paraphrasing*: the automated generation of syntactically diverse but semantically equivalent text [1]. By varying vocabulary and sentence structure while retaining the original malicious intent, attackers can render static signatures obsolete. This capability effectively collapses the "cost of quality," allowing low-resource adversaries to scale high-context spear phishing campaigns that were previously operationally expensive to produce [3].

2.1 The Scale of the Threat

The transition from human-crafted to AI-generated spam is no longer theoretical; it is the dominant reality of the current threat landscape. Hao et al. (2025) provide the first large-scale empirical measurement of this phenomenon, analyzing wild email traffic to quantify the prevalence of synthetic text. Their study estimates that 51% of spam emails are now generated using LLMs [2].

This saturation fundamentally invalidates the primary heuristic used by defenders for the last decade: reliance on linguistic imperfection. Historically, syntax errors and awkward phrasing were high-fidelity signals of fraud. Today, however, grammatical perfection is not a sign of legitimacy, but a baseline characteristic of automated generation.

Attackers leverage this fluency to scale psychological manipulation through the "V-Triad" framework. Heiding et al. (2024) demonstrate that LLMs excel at automating three specific persuasion tactics: *Visceral Triggers*, which use emotive language to induce immediate reactions like fear of account loss; *Verification Cues*, which mimic authoritative processes such as IT audits to build false trust; and *Vulnerability Exploitation*, which targets specific cognitive biases or user habits. When these psychological vectors are

combined with error-free text, phishing success rates, measured by click-throughs, surge to between 43% and 81%, vastly outperforming human-written baselines [3].

Furthermore, this proliferation introduces a secondary, long-term threat: *Generative Poisoning*. As AI-generated spam floods the open web, it pollutes the data reservoirs used to train future Large Language Models. This phenomenon, often termed "Model Collapse," creates a feedback loop where future defensive models may lose the ability to distinguish human from machine text, as the "human" baseline becomes inextricably contaminated by synthetic artifacts. Thus, the evasion attacks of today become the poisoning attacks of tomorrow.

2.2 Mechanisms of Evasion

To operationalize this polymorphism, attackers employ algorithmic evasion strategies designed to traverse the decision boundaries of classifiers.

2.2.1 Adversarial Paraphrasing

Cheng et al. (2025) define "Adversarial Paraphrasing" as a training-free framework where a secondary LLM acts as an optimizer against a detection model. Unlike standard paraphrasing, which prioritizes readability, this method optimizes for *invisibility*. By iteratively replacing tokens to strip away statistical artifacts, such as low perplexity, while strictly preserving semantic intent, attackers can traverse the decision boundaries of classifiers. This technique is devastatingly effective: Cheng et al. report that it reduces the True Positive Rate (TPR) of detectors by up to 98.96%, effectively rendering stylometric defenses fragile in adversarial settings [1].

2.2.2 Taxonomy of Perturbation

Hotoğlu et al. (2025) further deconstruct these evasion strategies into a systematic taxonomy of perturbation. At the *word level*, attackers utilize "Spam Weight" scoring functions to identify high-risk tokens (e.g., "urgent", "verify") and algorithmically swap them for contextually similar but lower-weighted synonyms. At the *character level*, adversaries employ homoglyph attacks, substituting Latin characters with visually identical Cyrillic or Greek alternatives to defeat tokenization; while the text appears unchanged to a human reader, the classifier perceives it as benign noise. Finally, at the *structural level*, LLMs are prompted to artificially inject "burstiness", altering sentence length and complexity to mimic human irregularity, thereby neutralizing stylometric features that rely on machine consistency [4].

3 The Defensive Paradigm: The Stylometric Trap

In response to the weaponization of perfect syntax, the defensive frontier has shifted from keyword detection to *Stylometry*, the statistical analysis of linguistic style [6]. The central hypothesis of this paradigm is that while Large Language Models (LLMs) can mimic human grammar, they cannot fully replicate human irregularity. They exhibit a "machine accent", a statistical rigidity born from their objective function to minimize perplexity, resulting in text that is mathematically distinguishable from organic human communication.

3.1 Stylometric Detection

Opara et al. (2025) operationalize this hypothesis by extracting 60 distinct stylometric features to identify synthetic phishing attempts. Their methodology moves beyond surface-level keywords to the structural "DNA" of the text, relying on the fundamental divergence in how humans and machines generate language.

Mathematically, LLMs are trained to minimize *Perplexity* (*PPL*), defined as the inverse probability of the test set normalized by the number of words (N):

$$PPL(W) = P(w_1, \dots, w_N)^{-\frac{1}{N}} \quad (1)$$

Consequently, AI-generated text exhibits a consistent, low-perplexity statistical signature, choosing the most probable "smooth" path through the language manifold. In contrast, human writing is characterized by high *Burstiness*, sudden, irregular spikes in perplexity caused by creative vocabulary and non-linear logic.

Opara et al. capture these dynamics through specific metrics: *Clause Density* measures syntactic complexity, where the uniform structures of LLMs contrast with the nested, fragmented syntax (high burstiness) of human writing. *Lexical Diversity* exploits the probabilistic nature of LLMs; because models default to high-probability tokens, they consistently exhibit lower vocabulary variance compared to the natural lexical chaos of human authors. This is augmented by *Imperative Verb Count*, revealing that AI-generated phishing employs a hyper-concentration of command verbs (e.g., "verify," "click") distinct from the neutral tone of standard AI outputs.

By training an XGBoost classifier on these features, Opara et al. achieved a detection accuracy of 96% and an Area Under the Curve (AUC) of 99% in distinguishing GPT-4 generated content from human text [6]. These metrics suggest that, in a vacuum, the "machine accent" is a highly detectable artifact.

3.2 Critical Analysis of Stylometry

While the performance of Opara's classifier appears robust, it reveals a fundamental fragility when placed in an adversarial context. The classifier succeeds primarily because

it detects the *default* output distribution of the LLM—text that is polite, formal, and statistically average.

However, as established in Section 2, this defense is static while the adversary is dynamic. The efficacy of stylometry relies on the attacker making no effort to obfuscate the source. If an attacker employs *Adversarial Paraphrasing* [1], prompting the model to "write with high burstiness" or injecting intentional typos, the stylometric features (e.g., uniform clause density) dissolve. Consequently, stylometry detects the *model*, not the *malicious intent*. It is a filter for "lazy" attackers, not sophisticated ones.

The gap between academic success and operational reality is stark. Opara et al. observed that while their custom XGBoost model succeeded, commercial filters failed catastrophically. In their experiments, Microsoft Outlook flagged only 4% of AI-generated phishing samples [6]. This discrepancy indicates that major providers are still optimized for the previous era of "broken" spam and have not yet integrated sufficient stylometric analysis to counter the wave of synthetic perfection.

4 The Failure of Human-AI Teaming

Given the demonstrated fragility of automated filters, a prevailing industry response has been to position the user as the "last line of defense," augmented by LLM-based decision support tools. This "Human-in-the-Loop" paradigm posits that an AI assistant can highlight suspicious elements (e.g., "This email contains urgent language"), allowing the human to make the final adjudication. However, Flores et al. [5] demonstrate that this approach often weakens, rather than strengthens, overall security posture due to a psychological phenomenon known as *Cognitive Offloading*.

4.1 The Paradox of Assistance

Flores et al. [5] conducted a controlled study (N=295) to measure the impact of LLM assistants on user detection accuracy. The results reveal a dangerous paradox: while assistants marginally improved the performance of novice users, they significantly degraded the accuracy of knowledgeable participants. This degradation is driven by *Automation Bias*, the tendency of humans to suspend critical judgment in favor of algorithmic advice.

The study found that when the LLM assistant provided an incorrect "Safe" verdict on a sophisticated phishing email, users were statistically likely to override their own correct suspicions and accept the AI's judgment. Instead of acting as a "second pair of eyes," the LLM induces a state of cognitive dependency.

4.2 The Recursive Vulnerability

This reliance creates a recursive vulnerability when paired with the adversarial attacks described in Section 2. If an attacker uses Cheng et al.'s [1] *Adversarial Paraphrasing* to

lower the "AI Score" of an email, the defensive LLM assistant will likely misclassify it as benign. Because the human user is engaging in cognitive offloading, they will interpret this silence as a confirmation of safety. Consequently, the "Human-in-the-Loop" does not provide defense-in-depth; rather, it creates a single point of failure where the human merely rubber-stamps the errors of the machine.

5 Future Outlook: From Form to Intent

The adversarial dynamics described above culminate in a defensive paradox: attackers can optimize linguistic "form" faster than defenders can characterize it. To escape this cycle, the security paradigm must pivot from analyzing *Internal Syntax*, the statistical properties of the text, to verifying *External Context*, the operational validity of the request.

Wang et al. [7] exemplify this shift with "SmishX," an agentic framework that abandons the binary classification of "Human vs. AI" in favor of automated forensic investigation. Instead of asking "Who wrote this?", the system asks "Is this true?" It employs Large Language Model Agents equipped with Chain-of-Thought (CoT) reasoning to autonomously investigate message claims against external reality. The system actively queries external data sources, cross-referencing URL redirect chains against claimed senders and validating domain ownership via WHOIS data to establish brand consistency. By anchoring detection in verifiable external facts rather than malleable stylistic features, this agentic approach achieved 98.8% accuracy, demonstrating resilience against the linguistic perturbations that render stylometric filters obsolete.

6 Conclusion

The integration of Large Language Models (LLMs) into the email ecosystem represents a fundamental shift in the economics of cybercrime. We have transitioned from a "Syntax Era," where defense relied on the attacker's inability to write well, to a "Semantic Era," where the attacker writes perfectly at scale. This essay has demonstrated that while stylometric detection (Opara et al.) currently offers a high-fidelity shield against generic AI text, it remains structurally fragile against the emerging class of adversarial paraphrasing attacks (Cheng et al.) [6, 1].

Furthermore, the strategy of offloading detection to human-AI teams is undermined by automation bias, where the improved capabilities of LLM assistants paradoxically lower the vigilance of expert users (Flores et al.) [5]. Consequently, the current defensive paradigm, which relies on identifying the *source* of the text, is becoming obsolete. The necessary evolution is toward *Intent Verification*: deploying agentic systems (Wang et al.) that autonomously validate the external claims of a message, thereby anchoring security in verifiable reality rather than manipulable style [7].

References

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Self-Evaluation

1. Essay Readability and Structure

I think the essay reads well and flows logically. I followed the required "paper-like" structure, ensuring I had a clear Abstract, Introduction, Body, and Conclusion. I prioritized tight, declarative sentences to avoid fluff (e.g., "In today's digital world"). I also tried to explain technical concepts intuitively, like the "Stylometric Trap" and "Adversarial Paraphrasing", before introducing specific taxonomies, so it is easier for the reader to digest the conflict between offense and defense. Finally, I double-checked that I included proper IEEE citations for every claim.

2. Topic and Understanding

For this assignment, I chose the *LLMs in Spam Filtering* topic (specifically Poisoning and Evasion). While I studied the mandatory papers (Opara et al., Flores et al.), I also read Hao et al. to quantify the scale of the threat and Cheng et al. to grasp the mechanics of adversarial evasion. I feel I have developed a strong, deep understanding of the material. I moved beyond simple definitions to explain the mathematical foundations of detection (minimizing Perplexity vs. Burstiness) and the mechanisms of "Spam Weight" optimization. I also addressed the full scope of the prompt by linking immediate evasion attacks to the long-term threat of "Generative Poisoning" (Model Collapse).

3. Grade Justification

I believe this paper merits at least a Pass because it meets all assignment requirements: correct formatting, page limits, mandatory citations, and the collaboration footnote. I clearly summarized the core contributions of Opara and Flores, providing necessary background for the reader.

Beyond a pass, I hope this essay is considered Excellent because I demonstrated critical synthesis and technical rigor rather than simple summarization. I provided specific mathematical depth by defining Perplexity and Burstiness to explain the intuition behind stylometric success. I also addressed the full scope of the assignment by connecting the saturation of AI spam to the long-term risk of "Generative Poisoning" and Model Collapse. Furthermore, I constructed an adversarial narrative that explicitly pitted the findings of Cheng against Opara to expose the fragility of current defenses, concluding with a concrete architectural proposal for Agentic Defense (Wang et al.) to move the discussion toward future solutions.